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Torsional coupling for electric hydraulic fracturing fluid pumps

Abstract

A system for hydraulically fracturing an underground formation in an oil or gas well, including a pump for pumping hydraulic fracturing fluid into the wellbore, the pump having a pump shaft, and an electric motor with a motor shaft mechanically attached to the pump to drive the pump. The system further includes a torsional coupling connecting the motor shaft to the pump shaft. The torsional coupling includes a motor component fixedly attached to the motor shaft and having motor coupling claws extending outwardly away from the motor shaft, and a pump component fixedly attached to the pump shaft of the pump and having pump coupling claws extending outwardly away from the pump shaft. The motor coupling claws engage with the pump coupling claws so that when the motor shaft and motor component rotate, such rotation causes the pump component and the pump shaft to rotate, thereby driving the pump.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/379,651 filed Jul. 19, 2021 titled “TORSIONAL COUPLING FOR ELECTRIC HYDRAULIC FRACTURING FLUID PUMPS,” which is now U.S. Pat. No. 11,549,346 issued Jan. 10, 2023, which is a continuation of U.S. patent application Ser. No. 15/581,625 filed Apr. 28, 2017 titled “TORSIONAL COUPLING FOR ELECTRIC HYDRAULIC FRACTURING FLUID PUMPS,” which is now U.S. Pat. No. 11,066,912 issued Jul. 20, 2021, which is a continuation of U.S. patent application Ser. No. 14/622,532 filed Feb. 13, 2015 titled “TORSIONAL COUPLING FOR ELECTRIC HYDRAULIC FRACTURING FLUID PUMPS,” which is now U.S. Pat. No. 9,650,879 issued May 16, 2017, which is a continuation-in-part of, and claims priority to and the benefit of, U.S. patent application Ser. No. 13/679,689 filed Nov. 16, 2012 titled “SYSTEM FOR PUMPING HYDRAULIC FRACTURING FLUID USING ELECTRIC PUMPS,” which is now U.S. Pat. No. 9,410,410 issued Aug. 9, 2016, the full disclosures of which are incorporated herein by reference for all intents and purposes.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) This technology relates to hydraulic fracturing in oil and gas wells. In particular, this technology relates to pumping fracturing fluid into an oil or gas well using pumps powered by electric motors.

2. Brief Description of Related Art

(2) Typically, motors are used at a well site to drive equipment. For example, diesel, gas, or electric motors might be used to drive pumps, blenders, or hydration units for carrying out hydraulic fracturing operations. Such motors are attached to the well site equipment by connecting the shaft of the motor to a shaft on the equipment, such a pump shaft for a pump, or a hydraulic motor shaft for a blender or a hydration unit. In order to compensate for misalignment between the motor and the equipment driven by the motor, a U-joint shaft is typically used. The U-joint shaft allows limited radial, angular, or even axial misalignment between the motor and the equipment, while still allowing mechanical communication between the shafts of the motor and the equipment to drive the equipment.

(3) Use of U-joint shafts, however, can be problematic in practice. For example, U-joint shafts introduce inefficiencies into the system, losing up to 10% or more of the energy that would otherwise be transmitted from the motor shaft to the equipment. Furthermore, a minimum of 3 degrees of offset can be required between the motor and the equipment in order for the U-joint shaft to function properly. This offset leads to the need for a longer shaft, which in turn leads to greater separation between the motor and the equipment. Such separation can be problematic in setup where space is limited, for example, where both the motor and a pump are mounted to a trailer or truck body.

SUMMARY OF THE INVENTION

(4) The present technology provides a system for hydraulically fracturing an underground formation in an oil or gas well. The system includes a pump for pumping hydraulic fracturing fluid

into the wellbore at high pressure so that the fluid passes from the wellbore into the formation and fractures the formation, the pump having a pump shaft that turns to activate the pump. The system further includes an electric motor with a motor shaft mechanically attached to the pump to drive the pump, and a torsional coupling connecting the motor shaft to the pump shaft. The torsional coupling has a motor component fixedly attached to the motor shaft of the electric motor and having motor coupling claws extending outwardly away from the motor shaft, and a pump component fixedly attached to the pump shaft of the pump and having pump coupling claws extending outwardly away from the pump shaft. The motor coupling claws engage with the pump coupling claws so that when the motor shaft and motor component rotate, such rotation causes the pump component and the pump shaft to rotate, thereby driving the pump.

(5) In some embodiments, the pump component or the motor component can further include elastomeric inserts positioned between the pump coupling claws or the motor coupling claws, respectively, to provide a buffer therebetween and to absorb movement and vibration in the torsional coupling. In addition, the motor coupling claws and the pump coupling claws can be spaced to allow radial misalignment, axial misalignment, or angular misalignment of the motor component and the pump component while still allowing engagement of the motor component and the pump component to transmit torque. Furthermore, the torsional coupling can further comprise a retainer cap attached to the motor component or the pump component to cover the interface therebetween and to prevent the ingress of debris or contaminants between the motor component and the pump component. The retainer cap can be removable from the torsional coupling to allow access to the inside of the coupling.

(6) In some embodiments, the motor component can have a tapered central bore for receiving the motor shaft. In addition, the pump and the motor can be mounted on separate but aligned weldments. Alternatively, the pump and the motor can be mounted on a single common weldment. Pump and motor mounted on single weldment for ease of alignment and stability.

(7) Another embodiment of the present technology provides a system for pumping hydraulic fracturing fluid into a wellbore. The system includes a pump having a pump shaft, an electric motor having a motor shaft mechanically attached to the pump to drive the pump, and a torsional coupling connecting the motor shaft to the pump shaft. The torsional coupling includes a motor component fixedly attached to the motor shaft and having motor coupling claws extending outwardly away from the motor shaft, and a pump component fixedly attached to the pump shaft and having pump coupling claws extending outwardly away from the pump shaft. The motor coupling claws engage with the pump coupling claws so that when the motor shaft and motor component rotate, such rotation causes the pump component and the pump shaft to rotate. In addition, the motor coupling claws and the pump coupling claws are spaced to allow radial misalignment, axial misalignment, or angular misalignment of the motor component and the pump component, while still allowing engagement of the motor component and the pump component to transmit torque.

(8) In some embodiments, the pump component or the motor component further include elastomeric inserts positioned between the pump coupling claws or the motor coupling claws, respectively, to provide a buffer therebetween and to absorb movement and vibration in the torsional coupling. In addition, the torsional coupling can further include a retainer cap attached to the motor component or the pump component to cover the interface therebetween and to prevent the ingress of debris or contaminants between the motor component and the pump component. The retainer cap can be removable from the torsional coupling to allow access to the inside of the coupling.

(9) In some embodiments, the motor component can have a tapered central bore for receiving the motor shaft. In addition, the pump and the motor can be mounted on separate but aligned weldments. Alternatively, the pump and the motor can be mounted on a single common weldment.

(10) Yet another embodiment of the present technology provides a system for conducting hydraulic fracturing operations in a well. The system includes hydraulic fracturing equipment, the hydraulic

fracturing equipment selected from the group consisting of a hydraulic fracturing pump, a hydraulic motor of a blender, and a hydraulic motor of a hydration unit, the hydraulic fracturing equipment having a hydraulic fracturing equipment shaft. The system further includes an electric motor with a motor shaft mechanically attached to the hydraulic fracturing equipment to drive the hydraulic fracturing equipment, and a torsional coupling connecting the motor shaft to the hydraulic fracturing equipment shaft. The torsional coupling includes a motor component fixedly attached to the motor shaft of the electric motor and having motor coupling claws extending outwardly away from the motor shaft, and a hydraulic fracturing equipment component fixedly attached to the hydraulic fracturing equipment shaft of the hydraulic fracturing equipment and having hydraulic fracturing equipment coupling claws extending outwardly away from the hydraulic fracturing equipment shaft. The motor coupling claws engage with the hydraulic fracturing equipment coupling claws so that when the motor shaft and motor component rotate, such rotation causes the hydraulic fracturing equipment component and the hydraulic fracturing equipment shaft to rotate, thereby driving the hydraulic fracturing equipment.

(11) In some embodiments, the hydraulic fracturing equipment component or the motor component can further include elastomeric inserts positioned between the hydraulic fracturing equipment coupling claws or the motor coupling claws, respectively, to provide a buffer therebetween and to absorb movement and vibration in the torsional coupling. In addition, the motor coupling claws and the hydraulic fracturing equipment coupling claws can be spaced to allow radial misalignment, axial misalignment, or angular misalignment of the motor component and the hydraulic fracturing equipment component while still allowing engagement of the motor component and the hydraulic fracturing equipment component to transmit torque.

(12) In some embodiments, the torsional coupling can further include a retainer cap attached to the motor component or the hydraulic fracturing equipment component to cover the interface therebetween and to prevent the ingress of debris or contaminants between the motor component and the hydraulic fracturing equipment component. In addition, the motor component can have a tapered central bore for receiving the motor shaft.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present technology will be better understood on reading the following detailed description of nonlimiting embodiments thereof, and on examining the accompanying drawing, in which:

(2) FIG. 1 is a schematic plan view of equipment used in a hydraulic fracturing operation, according to an embodiment of the present technology;

(3) FIG. 2A is a side view of a torsional coupling according to the present technology with the components of the coupling radially misaligned;

(4) FIG. 2B is a side view of a torsional coupling according to the present technology with the components of the coupling angularly misaligned;

(5) FIG. 2C is a side view of a torsional coupling according to the present technology with the components of the coupling axially misaligned;

(6) FIG. 3 is a perspective view of the torsional coupling with the components separated;

(7) FIG. 4 is an end view of the torsional coupling according to an embodiment of the present technology;

(8) FIG. 5 is a side cross-sectional view of the torsional coupling of FIG. 4 taken along the line 5-5 in FIG. 4;

(9) FIG. 6 is a side cross-sectional view of the torsional coupling according to an alternate embodiment of the present technology;

(10) FIG. 7A is a side view of a motor according to an embodiment of the present technology with

a part of the torsional coupling mounted to the motor shaft;

(11) FIG. 7B is a side cross-sectional view of the part of the torsional coupling shown in FIG. 7A, taken along line 7B-7B;

(12) FIG. 8 is a perspective view of a motor and torsional coupling according to an embodiment of the present technology;

(13) FIG. 9 is a side view of a motor and pump mounted to a single weldment;

(14) FIG. 10 is a schematic plan view of equipment used in a hydraulic fracturing operation, according to an alternate embodiment of the present technology;

(15) FIG. 11 is a left side view of equipment used to pump fracturing fluid into a well and mounted on a trailer, according to an embodiment of the present technology; and

(16) FIG. 12 is a right side view of the equipment and trailer shown in FIG. 3.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

(17) The foregoing aspects, features, and advantages of the present technology will be further appreciated when considered with reference to the following description of preferred embodiments and accompanying drawing, wherein like reference numerals represent like elements. In describing the preferred embodiments of the technology illustrated in the appended drawing, specific terminology will be used for the sake of clarity. However, the technology is not intended to be limited to the specific terms used, and it is to be understood that each specific term includes equivalents that operate in a similar manner to accomplish a similar purpose.

(18) FIG. 1 shows a plan view of equipment used in a hydraulic fracturing operation. Specifically, there is shown a plurality of pumps 10 mounted to vehicles 12, such as trailers (as shown, for example, in FIGS. 3 and 4). In the embodiment shown, the pumps 10 are powered by electric motors 14, which can also be mounted to the vehicles 12. The pumps 10 are fluidly connected to the wellhead 16 via the missile 18. As shown, the vehicles 12 can be positioned near enough to the missile 18 to connect fracturing fluid lines 20 between the pumps 10 and the missile 18. The missile 18 is then connected to the wellhead 16 and configured to deliver fracturing fluid provided by the pumps 10 to the wellhead 16. Although the vehicles 12 are shown in FIGS. 3 and 4 to be trailers, the vehicles could alternately be trucks, wherein the pumps motors 14, and other equipment are mounted directly to the truck.

(19) In some embodiments, each electric motor 14 can be an induction motor, and can be capable of delivering about 1500 horsepower (HP), 1750 HP, or more. Use of induction motors, and in particular three-phase induction motors, allows for increased power output compared to other types of electric motors, such as permanent magnet (PM) motors. This is because three-phase induction motors have nine poles (3 poles per phase) to boost the power factor of the motors. Conversely, PM motors are synchronous machines that are accordingly limited in speed and torque. This means that for a PM motor to match the power output of a three-phase induction motor, the PM motor must rotate very fast, which can lead to overheating and other problems.

(20) Each pump 10 can optionally be rated for about 2250 horsepower (HP) or more. In addition, the components of the system, including the pumps 10 and the electric motors 14, can be capable of operating during prolonged pumping operations, and in temperature in a range of about 0 degrees C. or less to about 55 degrees C. or more. In addition, each electric motor 14 can be equipped with a variable frequency drive (VFD) 15, and an A/C console, that controls the speed of the electric motor 14, and hence the speed of the pump 10.

(21) The VFDs 15 of the present technology can be discrete to each vehicle 12 and/or pump. Such a feature is advantageous because it allows for independent control of the pumps 10 and motors 14. Thus, if one pump 10 and/or motor 14 becomes incapacitated, the remaining pumps 10 and motors 14 on the vehicle 12 or in the fleet can continue to function, thereby adding redundancy and flexibility to the system. In addition, separate control of each pump 10 and/or motor 14 makes the system more scalable, because individual pumps 10 and/or motors 14 can be added to or removed from a site without modification to the VFDs 15.

(22) The electric motors **14** of the present technology can be designed to withstand an oilfield environment. Specifically, some pumps **10** can have a maximum continuous power output of about 1500 HP, 1750 HP, or more, and a maximum continuous torque of about 8750 ft-lb, 11,485 ft-lb, or more. Furthermore, electric motors **14** of the present technology can include class H insulation and high temperature ratings, such as about 1100 degrees C. or more. In some embodiments, the electric motor **14** can include a single shaft extension and hub for high tension radial loads, and a high strength **4340** alloy steel drive shaft, although other suitable materials can also be used.

(23) The VFD **15** can be designed to maximize the flexibility, robustness, serviceability, and reliability required by oilfield applications, such as hydraulic fracturing. For example, as far as hardware is concerned, the VFD **15** can include packaging receiving a high rating by the National Electrical Manufacturers Association (such as nema 1 packaging), and power semiconductor heat sinks having one or more thermal sensors monitored by a microprocessor to prevent semiconductor damage caused by excessive heat. Furthermore, with respect to control capabilities, the VFD **15** can provide complete monitoring and protection of drive internal operations while communicating with an operator via one or more user interfaces. For example, motor diagnostics can be performed frequently (e.g., on the application of power, or with each start), to prevent damage to a grounded or shorted electric motor **14**. The electric motor diagnostics can be disabled, if desired, when using, for example, a low impedance or high-speed electric motor.

(24) In some embodiments, the pump **10** can optionally be a 2250 HP triplex or quintuplex pump. The pump **10** can optionally be equipped with 4.5 inch diameter plungers that have an eight (8) inch stroke, although other size plungers can be used, depending on the preference of the operator. The pump **10** can further include additional features to increase its capacity, durability, and robustness, including, for example, a 6.353 to 1 gear reduction, autofrettaged steel or steel alloy fluid end, wing guided slush type valves, and rubber spring loaded packing. Alternately, pumps having slightly different specifications could be used. For example, the pump **10** could be equipped with 4 inch diameter plungers, and/or plungers having a ten (10) inch stroke.

(25) In certain embodiments of the invention, the electric motor **14** can be connected to the pump **10** via a torsional coupling **152**, of the type illustrated in FIGS. 2A-2C. Use of such a torsional coupling **152** is advantageous compared to use of, for example, a U-joint drive shaft to connect the motor **14** to the pump **10**, because the torsional coupling **152** is more efficient. For example, in a typically system, in which a pump is connected to and powered by a diesel motor, the pump may be connected to the diesel motor using a U-joint drive shaft. Such drive shafts typically require at least a 3 degree offset, and they may lose up to 10% or more energy due to inefficiencies. By replacing the U-joint drive shaft with a torsional coupling **152** in the system of the present technology, this inefficiency can be reduced to 1% or less. In addition, the torsional coupling **152** allows for a shorter driveshaft than the U-joint drive shaft, thereby requiring a smaller space. Such space savings is valuable in particular for trailer or truck mounted systems.

(26) The torsional coupling **152** of the present technology compensates for offset between a motor shaft and a pump shaft by allowing for some misalignment of the coupling components, while still maintaining an operative relationship between the components. For example, as shown in FIG. 2A, the pump component **154** of the coupling **152** can be radially offset from the motor component **156** of the coupling **152** by a radial distance R, and the two components **154**, **156** may still be engaged so that when the motor component **156** rotates it causes rotation of the pump component **154**. In fact, in some embodiments, the radial distance R can be up to 1.8 mm or more.

(27) Similarly, as shown in FIG. 2B, the pump component **154** can be angled relative to the motor component **156** of the coupling **152** at an angle θ , and the two components **154**, **156** may still be engaged. In some instances, the angle θ may be up to about 0.33 degrees. In addition, as shown in FIG. 2C, the pump component **154** can be axially separated from the motor component **156** by a distance S, and the two components **154**, **156** may still be engaged. In some embodiments, the components **154**, **156** can be axially separated by an axial distance S of up to 110 mm or more.

(28) Referring now to FIG. 3, there is shown an isometric view of the pump component 154 and the motor component 156 of the coupling 152. The pump component 154 includes a protrusion 158 extending perpendicularly outward toward the pump (not shown), and which has a bore 160 configured to receive the shaft with an interference fit so that the pump component 154 transmits torque to the shaft of the pump when the pump component 154 turns. The pump component 154 also includes pump coupling claws 162 that extend inwardly toward the motor component 156 of the coupling 152 when the coupling 152 is made up. The pump coupling claws 162 are spaced circumferentially around the pump component 154. In some embodiments, such as that shown in FIG. 3, there can be six pump coupling claws 162, but any appropriate number can be used.

(29) In addition to the above, the pump component 154 of the coupling 152 can include elastomeric inserts 164 surrounding at least a portion of the pump coupling claws 162 to provide a buffer between the pump coupling claws 162 of the pump component 154 and corresponding claws on the motor component 156. Such a buffer is advantageous to increase the ability of the coupling 152 to withstand shocks and vibrations associated with the use of heavy duty equipment such as hydraulic fracturing pumps. It is advantageous, when making up the coupling 152, to ensure that the components 154, 156 of the coupling are not mounted too far away from each other in an axial direction, so that the elastomeric inserts can transmit torque over the entire width of the inserts.

(30) Also shown in FIG. 3 is an isometric view of the motor component 156 according to an embodiment of the present technology. The motor component 156 includes a protrusion 166 extending perpendicularly outward toward the motor (not shown), and which has a bore 168. The bore 168 engages the shaft of the motor with an interference fit, so that the motor component 156 receives torque from the shaft of the motor. In some embodiments, the shaft may be tapered, as described in greater detail below. This taper helps, among other things, to properly set the depth of the motor shaft relative to the motor component 156 when making up the coupling 152. The interference fit of the pump shaft and the motor shaft into the pump and motor components 154, 156 of the coupling 152 can be achieved by heating the pump and motor components 154, 156 to, for example, about 250 degrees Fahrenheit, and installing the components on their respective shafts while hot. Thereafter, as the pump and motor components 154, 156 cool, the inner diameters of the bores 160, 168 in the pump and motor components 154, 156 decrease, thereby creating an interference fit between the pump and motor components 154, 156 and the pump and motor shafts, respectively.

(31) The motor component 156 also includes motor coupling claws 170 that extend inwardly toward the pump component 154 of the coupling 152 when the coupling 152 is made up. The motor coupling claws 170 are spaced circumferentially around the motor component 156 so as to correspond to voids between the pump coupling claws 162 and elastomeric inserts 164 when the coupling 152 is made up. In some embodiments, a retainer cap 172 can be included to cover the interface between the pump component 154 and the motor component 156, to protect, for example, the coupling 152 from the ingress of foreign objects or debris. The retainer cap 172 can be integral to the pump component 154 or it can be a separate piece that is fastened to the pump component 154.

(32) Thus, when the coupling 152 is made up, the motor shaft, which is inserted into the bore 168 of the motor component 156, can turn and transmit torque to the motor component 156 of the coupling 152. As the motor component 156 of the coupling 152 turns, the motor coupling teeth 170 transmit torque to the pump coupling teeth 162 through the elastomeric inserts 164. Such torque transmission in turn causes the pump component 154 of the coupling 152 to turn, which transmits torque to the pump shaft engaged with the bore 160 of the pump component 154. The transmission of torque through the coupling 152 occurs even if the motor component 156 and the pump component 154 are radially offset, positioned at an angle to one another, or separated by an axial distance, as shown in FIGS. 2A-2C.

(33) Referring now to FIG. 4, there is shown an end view of the coupling 152 looking from the

pump side of the coupling **152** toward the motor. In particular, there is shown the pump component **154** of the coupling **152**, including the protrusion **158** and the bore **160** for receiving the pump shaft. In the embodiment of FIG. **4**, the retainer cap **172** is a separate piece from the pump component **154**, and is attached to the pump component **154** with fasteners **174**. In this embodiment shown, the fasteners **174** are shown to be bolts, but any appropriate fasteners could be used. Provision of a removable retainer cap **172** can be advantageous because it allows easier access to the interior components of the coupling **152** for servicing or repair. For example, if an operator desires to replace the elastomeric inserts **164** within the coupling **152**, it need only remove the retainer cap **172**, after which it can easily replace the elastomeric inserts **164**.

(34) FIG. **5** shows a cross-sectional view of the coupling **152** of FIG. **3**, taken along line 5-5. As shown in FIG. **5**, the bore **168** in the protrusion **166** of the motor component **156** of the coupling **152** can be tapered from a smaller diameter at an inward side **176** of the motor component **156** (toward the pump component **154**) to a larger diameter at an outward side **178** of the motor component (toward the motor). The tapered diameter of the bore **168** corresponds to a similarly tapered end of the motor shaft, and helps with torque transmission and depth setting of the motor shaft relative to the coupling **152** when the coupling **152** is made up.

(35) FIG. **6** shows a cross-sectional view of the coupling **152** according to an alternate embodiment of the present technology, and including the motor shaft **180** and pump shaft **182**. In addition, in the view shown in FIG. **6**, there is shown the elastomeric inserts **164** in the coupling. Furthermore, the embodiment shown in FIG. **6** differs from that shown in FIG. **5** in that the retainer cap **172** is integral to the pump component **154** (as opposed to being a separate piece, as depicted in FIGS. **4** and **5**).

(36) FIG. **7A** shows the motor component **156** of the coupling **152** attached to a motor **14**. As can be seen, the motor shaft **180** extends outwardly from the motor **14** and into engagement with the motor component **156**. FIG. **7B** shows how the end of the motor shaft **180** is tapered so that it fits within the tapered bore **168** of the motor component **156**. With the motor shaft **180** thus engaged with the motor component **156**, the motor shaft **180** transmits torque to the motor component **156** as the shaft **180** turns, thereby turning the motor component **156** as well.

(37) Referring now to FIG. **8**, there is shown a motor **14** according to an embodiment of the present invention, and a coupling **152**. There is also shown a protective cage **184** surrounding the coupling **152**. The protective cage provides the advantage of protecting the coupling **152** from damage. In addition, the protective cage **184** can have a removable panel **185**, or can otherwise be removable, to allow access to the coupling for repair and maintenance.

(38) The coupling **152** of the present technology can be built out of any suitable materials, including composite materials, and is designed to allow for high torsional forces. For example, the torque capacity of the coupling could be up to about 450,000 lb-in. In addition, when the motor, pump, and associated coupling **152** are mounted to a trailer, truck, skid, or other equipment, various sized shim plates can be used to allow for more precise positioning of the equipment, thereby leading to appropriate alignment of the shafts and coupling components. Support brackets may also be provided to fix the motor and the pump in place relative to the trailer, truck, skid, or other equipment, thereby helping to maintain such alignment.

(39) Furthermore, the pump and motor mounting may be separate weldments, or, as shown in FIG. **9**, they may alternatively be a combined single weldment **187**. If they are a single weldment **187**, the mounting faces can be machined, leveled, and planar to each other to increase the accuracy of alignment. Attaching the motor **14** and pump **10** to a single weldment **187** can be advantageous because it can improve alignment of the components, which can lead to reduced torsional stresses in the coupling. Mounting the motor **14** and pump **10** to a single weldment **187** also helps to ensure that during transport or operation, the motor **14** and pump are moved together, so that alignment of the coupling halves can be better maintained. In embodiments using separate weldments, the motor **14** can move independently of the pump **10**, thereby causing a misalignment of the components,

and possible damage to the coupling. In addition, the separate weldments can have a greater tendency to warp, requiring additional effort to get the alignment in the acceptable range.

(40) Use of the coupling **152** complements the combination of a triplex, plunger pump, and an electric motor **14**, because such a pump **10** and motor **14** are torsionally compatible. In other words, embodiments using this pump **10** and motor **14** are substantially free of serious torsional vibration, and vibration levels in the pump input shaft and in the coupling **152** are, as a result, kept within acceptable levels.

(41) For example, experiments testing the vibration of the system of the present technology have indicated that, in certain embodiments, the motor shaft vibratory stress can be about 14% of the allowable limit in the industry. In addition, the coupling maximum combined order torque can be about 24% of the allowable industry limit, vibratory torque can be about 21% of the allowable industry limit, and power loss can be about 25% of the allowable industry limit. Furthermore, the gearbox maximum combined order torque can be about 89% of the standard industry recommendations, and vibratory torque can be about 47% of standard industry recommendations, while the fracturing pump input shaft combined order vibratory stress can be about 68% of the recommended limit.

(42) The coupling **152** can further be used to connect the motor shaft **180** with other equipment besides a pump. For example, the coupling **152** can be used to connect the motor to a hydraulic drive powering multiple hydraulic motors in a hydration unit, or associated with blender equipment. In any of these applications, it is advantageous to provide a protective cage around the coupling **152**, and also to provide an easy access panel in the protective cage to provide access to the coupling **152**.

(43) In addition to the above, certain embodiments of the present technology can optionally include a skid (not shown) for supporting some or all of the above-described equipment. For example, the skid can support the electric motor **14** and the pump **10**. In addition, the skid can support the VFD **15**. Structurally, the skid can be constructed of heavy-duty longitudinal beams and cross-members made of an appropriate material, such as, for example, steel. The skid can further include heavy-duty lifting lugs, or eyes, that can optionally be of sufficient strength to allow the skid to be lifted at a single lift point. It is to be understood, however, that a skid is not necessary for use and operation of the technology, and the mounting of the equipment directly to a vehicle **12** without a skid can be advantageous because it enables quick transport of the equipment from place to place, and increased mobility of the pumping system.

(44) Referring back to FIG. **1**, also included in the equipment is a plurality of electric generators **22** that are connected to, and provide power to, the electric motors **14** on the vehicles **12**. To accomplish this, the electric generators **22** can be connected to the electric motors **14** by power lines (not shown). The electric generators **22** can be connected to the electric motors **14** via power distribution panels (not shown). In certain embodiments, the electric generators **22** can be powered by natural gas. For example, the generators can be powered by liquefied natural gas. The liquefied natural gas can be converted into a gaseous form in a vaporizer prior to use in the generators. The use of natural gas to power the electric generators **22** can be advantageous because above ground natural gas vessels **24** can already be placed on site in a field that produces gas in sufficient quantities. Thus, a portion of this natural gas can be used to power the electric generators **22**, thereby reducing or eliminating the need to import fuel from offsite. If desired by an operator, the electric generators **22** can optionally be natural gas turbine generators, such as those shown in FIG. **10**. The generators can run on any appropriate type of fuel, including liquefied natural gas (LNG).

(45) FIG. **1** also shows equipment for transporting and combining the components of the hydraulic fracturing fluid used in the system of the present technology. In many wells, the fracturing fluid contains a mixture of water, sand or other proppant, acid, and other chemicals. Examples of fracturing fluid components include acid, anti-bacterial agents, clay stabilizers, corrosion inhibitors, friction reducers, gelling agents, iron control agents, pH adjusting agents, scale inhibitors, and

surfactants. Historically, diesel has at times been used as a substitute for water in cold environments, or where a formation to be fractured is water sensitive, such as, for example, clay. The use of diesel, however, has been phased out over time because of price, and the development of newer, better technologies.

(46) In FIG. 1, there are specifically shown sand transporting vehicles **26**, an acid transporting vehicle **28**, vehicles for transporting other chemicals **30**, and a vehicle carrying a hydration unit **32**. Also shown are fracturing fluid blenders **34**, which can be configured to mix and blend the components of the hydraulic fracturing fluid, and to supply the hydraulic fracturing fluid to the pumps **10**. In the case of liquid components, such as water, acids, and at least some chemicals, the components can be supplied to the blenders **34** via fluid lines (not shown) from the respective component vehicles, or from the hydration unit **32**. In the case of solid components, such as sand, the component can be delivered to the blender **34** by a conveyor belt **38**. The water can be supplied to the hydration unit **32** from, for example, water tanks **36** onsite. Alternately, the water can be provided by water trucks. Furthermore, water can be provided directly from the water tanks **36** or water trucks to the blender **34**, without first passing through the hydration unit **32**.

(47) In certain embodiments of the technology, the hydration units **32** and blenders **34** can be powered by electric motors. For example, the blenders **34** can be powered by more than one motor, including motors having 600 horsepower or more, and motors having 1150 horsepower or more. The hydration units **32** can be powered by electric motors of 600 horsepower or more. In addition, in some embodiments, the hydration units **32** can each have up to five (5) chemical additive pumps, and a 200 bbl steel hydration tank.

(48) Pump control and data monitoring equipment **40** can be mounted on a control vehicle **42**, and connected to the pumps **10**, electric motors **14**, blenders **34**, and other downhole sensors and tools (not shown) to provide information to an operator, and to allow the operator to control different parameters of the fracturing operation. For example, the pump control and data monitoring equipment **40** can include an A/C console that controls the VFD **15**, and thus the speed of the electric motor **14** and the pump **10**. Other pump control and data monitoring equipment can include pump throttles, a pump VFD fault indicator with a reset, a general fault indicator with a reset, a main estop, a programmable logic controller for local control, and a graphics panel. The graphics panel can include, for example, a touchscreen interface.

(49) Referring now to FIG. 10, there is shown an alternate embodiment of the present technology. Specifically, there is shown a plurality of pumps **110** which, in this embodiment, are mounted to pump trailers **112**. As shown, the pumps **110** can optionally be loaded two to a trailer **112**, thereby minimizing the number of trailers needed to place the requisite number of pumps at a site. The ability to load two pumps **110** on one trailer **112** is possible because of the relatively light weight of the electric powered pumps **110** compared to other known pumps, such as diesel pumps. In the embodiment shown, the pumps **110** are powered by electric motors **114**, which can also be mounted to the pump trailers **112**. Furthermore, each electric motor **114** can be equipped with a VFD **115**, and an A/C console, that controls the speed of the motor **114**, and hence the speed of the pumps **110**.

(50) The VFDs **115** shown in FIG. 10 can be discrete to each pump trailer **112** and/or pump **110**. Such a feature is advantageous because it allows for independent control of the pumps **110** and motors **114**. Thus, if one pump **110** and/or motor **114** becomes incapacitated, the remaining pumps **110** and motors **114** on the pump trailers **112** or in the fleet can continue to function, thereby adding redundancy and flexibility to the system. In addition, separate control of each pump **110** and/or motor **114** makes the system more scalable, because individual pumps **110** and/or motors **114** can be added to or removed from a site without modification to the VFDs **115**.

(51) In addition to the above, and still referring to FIG. 10, the system can optionally include a skid (not shown) for supporting some or all of the above-described equipment. For example, the skid can support the electric motors **114** and the pumps **110**. In addition, the skid can support the VFD

115. Structurally, the skid can be constructed of heavy-duty longitudinal beams and cross-members made of an appropriate material, such as, for example, steel. The skid can further include heavy-duty lifting lugs, or eyes, that can optionally be of sufficient strength to allow the skid to be lifted at a single lift point. It is to be understood that a skid is not necessary for use and operation of the technology and the mounting of the equipment directly to a trailer **112** may be advantageous because it enables quick transport of the equipment from place to place, and increased mobility of the pumping system, as discussed above.

(52) The pumps **110** are fluidly connected to a wellhead **116** via a missile **118**. As shown, the pump trailers **112** can be positioned near enough to the missile **118** to connect fracturing fluid lines **120** between the pumps **110** and the missile **118**. The missile **118** is then connected to the wellhead **116** and configured to deliver fracturing fluid provided by the pumps **110** to the wellhead **116**.

(53) This embodiment also includes a plurality of turbine generators **122** that are connected to, and provide power to, the electric motors **114** on the pump trailers **112**. To accomplish this, the turbine generators **122** can be connected to the electric motors **114** by power lines (not shown). The turbine generators **122** can be connected to the electric motors **114** via power distribution panels (not shown). In certain embodiments, the turbine generators **122** can be powered by natural gas, similar to the electric generators **22** discussed above in reference to the embodiment of FIG. **1**. Also included are control units **144** for the turbine generators **122**. The control units **144** can be connected to the turbine generators **122** in such a way that each turbine generator **122** is separately controlled. This provides redundancy and flexibility to the system, so that if one turbine generator **122** is taken off line (e.g., for repair or maintenance), the other turbine generators **122** can continue to function.

(54) The embodiment of FIG. **10** can include other equipment similar to that discussed above. For example, FIG. **10** shows sand transporting vehicles **126**, acid transporting vehicles **128**, other chemical transporting vehicles **130**, hydration unit **132**, blenders **134**, water tanks **136**, conveyor belts **138**, and pump control and data monitoring equipment **140** mounted on a control vehicle **142**. The function and specifications of each of these is similar to corresponding elements shown in FIG. **1**.

(55) Use of pumps **10**, **110** powered by electric motors **14**, **114** and natural gas powered electric generators **22** (or turbine generators **122**) to pump fracturing fluid into a well is advantageous over known systems for many different reasons. For example, the equipment (e.g. pumps, electric motors, and generators) is lighter than the diesel pumps commonly used in the industry. The lighter weight of the equipment allows loading of the equipment directly onto a truck body or trailer. Where the equipment is attached to a skid, as described above, the skid itself can be lifted on the truck body, along with all the equipment attached to the skid. Furthermore, and as shown in FIGS. **11** and **12**, trailers **112** can be used to transport the pumps **110** and electric motors **114**, with two or more pumps **110** carried on a single trailer **112**. Thus, the same number of pumps **110** can be transported on fewer trailers **112**. Known diesel pumps, in contrast, cannot be transported directly on a truck body or two on a trailer, but must be transported individually on trailers because of the great weight of the pumps.

(56) The ability to transfer the equipment of the present technology directly on a truck body or two to a trailer increases efficiency and lowers cost. In addition, by eliminating or reducing the number of trailers to carry the equipment, the equipment can be delivered to sites having a restricted amount of space, and can be carried to and away from worksites with less damage to the surrounding environment. Another reason that the electric powered pump system of the present technology is advantageous is that it runs on natural gas. Thus, the fuel is lower cost, the components of the system require less maintenance, and emissions are lower, so that potentially negative impacts on the environment are reduced.

(57) More detailed side views of the trailers **112**, having various system components mounted thereon, are shown in FIGS. **11** and **12**, which show left and right side views of a trailer **112**,

respectively. As can be seen, the trailer **112** can be configured to carry pumps **110**, electric motors **114** and a VFD **115**. Thus configured, the motors **114** and pumps **110** can be operated and controlled while mounted to the trailers **112**. This provides advantages such as increased mobility of the system. For example, if the equipment needs to be moved to a different site, or to a repair facility, the trailer can simply be towed to the new site or facility without the need to first load the equipment onto a trailer or truck, which can be a difficult and hazardous endeavor. This is a clear benefit over other systems, wherein motors and pumps are attached to skids that are delivered to a site and placed on the ground.

(58) In order to provide a system wherein the pumps **110**, motors **114**, and VFDs **115** remain trailer mounted, certain improvements can be made to the trailers **112**. For example, a third axle **146** can be added to increase the load capacity of the trailer and add stability. Additional supports and cross members **148** can be added to support the motors' torque. In addition, the neck **149** of the trailer can be modified by adding an outer rib **150** to further strengthen the neck **149**. The trailer can also include specially designed mounts **152** for the VFD **115** that allow the trailer to move independently of the VFD **115**, as well as specially designed cable trays for running cables on the trailer **112**. Although the VFD **115** is shown attached to the trailer in the embodiment of FIGS. **11** and **12**, it could alternately be located elsewhere on the site, and not mounted to the trailer **112**.

(59) In practice, a hydraulic fracturing operation can be carried out according to the following process. First, the water, sand, and other components are blended to form a fracturing fluid, which is pumped down the well by the electric-powered pumps. Typically, the well is designed so that the fracturing fluid can exit the wellbore at a desired location and pass into the surrounding formation. For example, in some embodiments the wellbore can have perforations that allow the fluid to pass from the wellbore into the formation. In other embodiments, the wellbore can include an openable sleeve, or the well can be open hole. The fracturing fluid can be pumped into the wellbore at a high enough pressure that the fracturing fluid cracks the formation, and enters into the cracks. Once inside the cracks, the sand, or other proppants in the mixture, wedges in the cracks, and holds the cracks open.

(60) Using the pump control and data monitoring equipment **40**, **140** the operator can monitor, gauge, and manipulate parameters of the operation, such as pressures, and volumes of fluids and proppants entering and exiting the well. For example, the operator can increase or decrease the ratio of sand to water as the fracturing process progresses and circumstances change.

(61) This process of injecting fracturing fluid into the wellbore can be carried out continuously, or repeated multiple times in stages, until the fracturing of the formation is optimized. Optionally, the wellbore can be temporarily plugged between each stage to maintain pressure, and increase fracturing in the formation. Generally, the proppant is inserted into the cracks formed in the formation by the fracturing, and left in place in the formation to prop open the cracks and allow oil or gas to flow into the wellbore.

(62) While the technology has been shown or described in only some of its forms, it should be apparent to those skilled in the art that it is not so limited, but is susceptible to various changes without departing from the scope of the technology. Furthermore, it is to be understood that the above disclosed embodiments are merely illustrative of the principles and applications of the present technology. Accordingly, numerous modifications can be made to the illustrative embodiments and other arrangements can be devised without departing from the spirit and scope of the present technology as defined by the appended claims.

Claims

1. A system for hydraulically fracturing an underground formation in an oil or gas well, the system comprising: a pump for pumping hydraulic fracturing fluid into a wellbore, the pump having a pump shaft that turns to activate the pump; an electric motor with a motor shaft to drive the pump

via the pump shaft; a variable frequency drive and an alternating current console to control the speed of the electric motor to protect against overheating; and a torsional coupling connecting the electric motor to the pump shaft, the torsional coupling comprising: a motor component coupled to the motor shaft of the electric motor; and a pump component coupled to the pump shaft of the pump; the motor component engaged with the pump component to transmit power from the electric motor to the pump when the motor shaft and the motor component rotate, the motor component contacting the pump component so that the pump component and the pump shaft rotate, thereby driving the pump; and the system further comprising buffers between the motor component and the pump component.

2. The system of claim 1, wherein the motor component further comprises motor coupling claws and the pump component further comprises pump coupling claws, and wherein the motor coupling claws and the pump coupling claws are spaced to allow at least one of radial, axial, or angular misalignment of the motor component and the pump component while still allowing engagement of the motor component and the pump component to transmit torque.

3. The system of claim 1, wherein the torsional coupling further comprises a retainer cap attached to at least one of the motor component or the pump component to cover the interface therebetween and to prevent ingress of debris or contaminants between the motor component and the pump component.

4. The system of claim 1, wherein the motor component has a tapered central bore for receiving the motor shaft.

5. The system of claim 1, wherein the pump and the motor are mounted on separate but aligned weldments.

6. The system of claim 1, wherein the pump and the motor are mounted on a single common weldment.

7. The system of claim 1, further comprising a variable frequency drive monitoring operation of the electric motor.

8. A system for pumping hydraulic fracturing fluid into a wellbore, the system comprising: a pump having a pump shaft; an electric motor having a motor shaft, the motor shaft coupling to the pump to drive the pump; a variable frequency drive and an alternating current console to control the speed of the electric motor to protect against overheating; and a torsional coupling for transmitting energy from the electric motor to the pump, the torsional coupling comprising: a motor component coupled to the motor shaft; a pump component coupled to the pump shaft; the motor component being coupled to the pump component to transmit rotation of the electric motor to the pump, wherein the motor component and the pump component are positioned to enable radial, axial, or angular misalignment when the motor component and the pump component are engaged; and the system further comprising buffers between the motor component and the pump component.

9. The system of claim 8, wherein the torsional coupling further comprises a retainer cap attached to at least one of the motor component or the pump component to cover the interface therebetween and to prevent ingress of debris or contaminants between the motor component and the pump component.

10. The system of claim 8, wherein the motor component has a tapered central bore for receiving the motor shaft.

11. The system of claim 8, wherein the pump and the motor are mounted on separate but aligned weldments.

12. The system of claim 8, further comprising a variable frequency drive monitoring operation of the electric motor.

13. A system for conducting hydraulic fracturing operations in a well, comprising: hydraulic fracturing equipment having a hydraulic fracturing equipment shaft; an electric motor with a motor shaft coupled to and driving the hydraulic fracturing equipment; a variable frequency drive and an alternating current console to control the speed of the electric motor to protect against overheating;

and a torsional coupling connecting the motor shaft to the hydraulic fracturing equipment shaft, the torsional coupling comprising: a motor component coupled to the motor shaft of the electric motor; and a hydraulic fracturing equipment component coupled to the hydraulic fracturing equipment shaft of the hydraulic fracturing equipment, the motor component engaging the hydraulic fracturing equipment component to drive operation of the hydraulic fracturing equipment when the electric motor shaft rotates; and the system further comprising buffers between the hydraulic fracturing equipment component and the motor component.

14. The system of claim 13, wherein the motor component further comprises motor coupling claws and the hydraulic fracturing equipment component further comprises hydraulic fracturing equipment claws, and wherein the motor coupling claws and the hydraulic fracturing equipment claws are spaced to allow at least one of radial, axial, or angular misalignment of the motor component and the hydraulic fracturing equipment component while still allowing engagement of the motor component and the hydraulic fracturing equipment component to transmit torque.

15. The system of claim 13, wherein the torsional coupling further comprises a retainer cap attached to at least one of the motor component or the hydraulic fracturing equipment component to cover the interface therebetween and to prevent ingress of debris or contaminants between the motor component and the hydraulic fracturing equipment component.

16. The system of claim 13, further comprising a variable frequency drive monitoring operation of the electric motor.
